

Neil's Algebra 1 — Tutor Math Packet

Concept teaching + practice problems with answer keys for Chapters 6–9 (Q4 exam scope)

How this packet is organized. Each chapter has three parts: (1) **Concepts & rules** — what the student needs to know, with key formulas highlighted; (2) **Common mistakes** — the specific errors most middle-schoolers make so you know what to watch for; (3) **Practice problems** — 8–12 graduated problems with answers in the side column for quick checking.

Suggested use. Spend ~20 min on the Concept page (teach / review with examples), then ~25 min on Practice (let Neil work, intervene only when stuck). The answer column lets you scan-check his work without flipping pages.

Pacing for the next 4 weeks (Phase 1, through May 20):

- Week 1 (Apr 27–May 3): **Chapter 6** — Exponents
- Week 2 (May 4–10): **Chapter 7** — Polynomials & Factoring (the prereq for Ch 8)
- Week 3 (May 11–17): **Chapter 8** — Quadratics + **Chapter 9** — Radicals
- Week 4 (May 18–20): Mock Q4 exam + final review

Chapter 6 — Exponents & Exponential Functions

Concepts & Rules

Exponent rules (must be automatic):

Product of powers: $x^a \cdot x^b = x^{a+b}$

Quotient of powers: $x^a / x^b = x^{a-b}$

Power of a power: $(x^a)^b = x^{ab}$

Power of a product: $(xy)^a = x^a \cdot y^a$

Zero exponent: $x^0 = 1$ (for $x \neq 0$)

Negative exponent: $x^{-a} = 1 / x^a$

Scientific notation:

A number written as $a \times 10^n$, where $1 \leq |a| < 10$. Move the decimal: each move left = +1 to exponent; each move right = -1.

Example: $0.00045 = 4.5 \times 10^{-4}$; $23,500,000 = 2.35 \times 10^7$

Exponential growth and decay:

Growth: $y = a(1 + r)^t$ (rate r as a decimal, t in time units)

Decay: $y = a(1 - r)^t$

a = starting value; r = growth/decay rate; t = time elapsed.

Common Mistakes

- Trying to combine bases that aren't the same (e.g., $x^2 \cdot y^3$ stays as is).
- Distributing wrongly: $(x + y)^2 \neq x^2 + y^2$. (FOIL it: $x^2 + 2xy + y^2$.)
- Negative exponent $\leftrightarrow$ negative answer. $x^{-2} = 1/x^2$, NOT $-x^2$.

- Forgetting that anything to the zero power is 1, not 0.
- Sign of the exponent in scientific notation: small numbers (< 1) have NEGATIVE exponents.

Practice Problems

Chapter 7 — Polynomials & Factoring

Concepts & Rules

Operations on polynomials:

- Add/subtract: combine LIKE TERMS only (same variable AND same exponent).
- Multiply binomials: FOIL — First, Outer, Inner, Last.

Example: $(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$

Factoring (the toolkit, in priority order):

1. Always pull out the GCF first.

2. Difference of squares: $a^2 - b^2 = (a + b)(a - b)$

3. Perfect-square trinomial: $a^2 \pm 2ab + b^2 = (a \pm b)^2$

4. $x^2 + bx + c$: Find two numbers with product c , sum b .

5. $ax^2 + bx + c$ ($a > 1$): AC method — find two numbers with product $a \cdot c$, sum b . Split the middle term, then group.

The AC method, step-by-step (Neil's most common stuck point):

Example: factor $6x^2 + 11x - 10$. $AC = 6 \cdot (-10) = -60$. Need two numbers: product -60 , sum $11 \rightarrow +15$ and -4 . Split: $6x^2 + 15x - 4x - 10$. Group: $3x(2x + 5) - 2(2x + 5) = (3x - 2)(2x + 5)$.

Common Mistakes

- Skipping the GCF check first. Always check.
- Sign errors when factoring $x^2 + bx + c$. Two numbers with product c , sum b — both signs depend on c and b .
- Confusing perfect square trinomial with general factoring (the middle term must be exactly $2ab$).
- Stopping too early. After GCF, the remaining factor may still factor further (e.g., $3x^3 - 12x \rightarrow 3x(x^2 - 4) \rightarrow 3x(x+2)(x-2)$).
- In FOIL, forgetting the middle terms or the sign on the last term.

Practice Problems

Chapter 8 — Quadratic Functions & Equations

This is the chapter Neil bombed on April 23. Most important chapter for Q4. The teacher's actual test had 7 distinct topic blocks — covered below.

Concepts & Rules

The three forms of a quadratic:

Standard form: $y = ax^2 + bx + c$

Vertex form: $y = a(x - h)^2 + k$ vertex (h, k)

Factored form: $y = a(x - r_1)(x - r_2)$ roots r_1, r_2

Graphing quadratics (the most common Q1 of any quadratics test):

1. Find axis of symmetry: $x = -b / (2a)$
2. Plug it into the equation to find the vertex's y-coordinate.
3. Build a small table around the axis (2 x-values left, 2 right).
4. Plot. Direction: $a > 0$ opens up, $a < 0$ opens down. Range: $y \geq \text{vertex-y}$ if up, $y \leq \text{vertex-y}$ if down. Domain is always all real numbers.

Vertex form & transformations from $v = x^2$:

$$y = a(x - h)^2 + k$$

- h: shifts horizontally — RIGHT if $h > 0$, LEFT if $h < 0$ (sign flips!)
- k: shifts vertically — UP if $k > 0$, DOWN if $k < 0$
- a: vertical stretch ($|a| > 1$) or compression ($|a| < 1$); negative a flips over x-axis

Discriminant — counting solutions without solving:

$$D = b^2 - 4ac$$

- $D > 0$: two real solutions (parabola crosses x-axis twice)
- $D = 0$: one real solution (parabola touches x-axis at vertex)
- $D < 0$: no real solutions (parabola never touches x-axis)

Solving quadratics — pick the right method:

1. **Factoring** — fastest if it factors. Set = 0, factor, zero-product property.
2. **Square roots** — when there's no x-term. Isolate x^2 , then $\pm \sqrt{}$.
3. **Completing the square** — when $a = 1$ and b is even. Move c, add $(b/2)^2$ to both sides, factor as $(x + b/2)^2$.
4. **Quadratic formula** — always works:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Word problems — projectile motion (Earth. feet. seconds):

$$h(t) = -16t^2 + v_0 t + h_0$$

- Max height: at $t = -b/(2a)$ (axis of symmetry); plug in to find h.
- Hits ground: solve $h(t) = 0$ (take positive t).

Linear vs. quadratic models from a data table:

- If first differences in y are CONSTANT → linear ($y = mx + b$).

- If first differences change but second differences are constant → quadratic. The constant second difference equals $2a$.

Common Mistakes

- Sign of h in vertex form. $y = (x + 4)^2 = (x - (-4))^2$, so $h = -4 \rightarrow$ shift LEFT 4 (not right).
- Forgetting the $\pm$ in the square-roots method. Always two solutions unless one of them is impossible.
- Quadratic formula: dropping the negative on $-b$, or losing parentheses around $(b^2 - 4ac)$.
- Discriminant: forgetting that c can be negative, which flips the sign of $-4ac$.
- Word problems: not rejecting impossible negative answers (e.g., width = -5).
- Confusing axis of symmetry $x = h$ with the vertex (which is (h, k)).

Practice Problems

Chapter 9 — Radicals, Pythagorean Theorem & Distance

Concepts & Rules

Simplifying square roots:

Pull out perfect-square factors: $\sqrt{48} = \sqrt{(16 \cdot 3)} = 4\sqrt{3}$

Memorize perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

Operations:

Multiplication: $\sqrt{a} \cdot \sqrt{b} = \sqrt{(ab)}$

Division: $\sqrt{a} / \sqrt{b} = \sqrt{(a/b)}$

Addition (LIKE radicals only): $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$

$3\sqrt{2} + 5\sqrt{3}$ cannot be simplified (different radicands).

Rationalizing the denominator:

Multiply numerator and denominator by the radical that makes the denominator rational.

Example: $4 / \sqrt{3} = (4 \cdot \sqrt{3}) / (\sqrt{3} \cdot \sqrt{3}) = 4\sqrt{3} / 3$

Pythagorean theorem (right triangles only):

$a^2 + b^2 = c^2$ (c = hypotenuse, the side opposite the right angle)

Distance and midpoint:

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$

Common Mistakes

- Not simplifying the radical fully. $\sqrt{72} \rightarrow 6\sqrt{2}$, not $\sqrt{72}$.
- Adding radicals with different radicands. $\sqrt{2} + \sqrt{3} \neq \sqrt{5}$.
- Pythagorean: solving for a leg vs. the hypotenuse. If finding a leg, rearrange to $a^2 = c^2 - b^2$.
- Distance formula: forgetting to square the differences before summing.
- Sign errors with negative coordinates inside the parentheses.

Practice Problems
